

Registration no = 2025/CS/079

Index no = ~~202~~ 25000799

(1)

1) $256_{10} = 100000000_2$

$$\begin{array}{r} 2 \overline{) 256} \\ 2 \overline{) 128} - 0 \\ 2 \overline{) 64} - 0 \\ 2 \overline{) 32} - 0 \\ 2 \overline{) 16} - 0 \\ 2 \overline{) 8} - 0 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 1 - 0 \end{array}$$

2) $354_{10} = 101100010_2$

$$\begin{array}{r} 2 \overline{) 354} \\ 2 \overline{) 177} - 0 \\ 2 \overline{) 88} - 1 \\ 2 \overline{) 44} - 0 \\ 2 \overline{) 22} - 0 \\ 2 \overline{) 11} - 0 \\ 2 \overline{) 5} - 1 \\ 2 \overline{) 2} - 1 \\ 1 - 0 \end{array}$$

3) $29_{10} = 11101_2$

$$\begin{array}{r} 2 \overline{) 29} \\ 2 \overline{) 14} - 1 \\ 2 \overline{) 7} - 0 \\ 2 \overline{) 3} - 1 \\ 1 - 1 \end{array}$$

4) $43_{10} = 101011_2$

$$\begin{array}{r} 2 \overline{) 43} \\ 2 \overline{) 21} - 1 \\ 2 \overline{) 10} - 1 \\ 2 \overline{) 5} - 0 \\ 2 \overline{) 2} - 1 \\ 1 - 0 \end{array}$$

5) $91_{10} = 1011011_2$

$$\begin{array}{r} 2 \overline{) 91} \\ 2 \overline{) 45} - 1 \\ 2 \overline{) 22} - 1 \\ 2 \overline{) 11} - 0 \\ 2 \overline{) 5} - 1 \\ 2 \overline{) 2} - 1 \\ 1 - 0 \end{array}$$

(2)

1) $111011_2 = 59_{10}$
 $1 + 2 + 8 + 16 + 32$

2) $1000101_2 = 38_{10}$
 $1 + 4 + 16 + 256$

3) $1000000_2 = 128_{10}$

4) $11011011_2 = 438_{10}$
 $2 + 4 + 16 + 32 + 128 + 256$

5) $110000100_2 = 388_{10}$
 $4 + 128 + 256$

(3)

1) $34_{10} = 42_8$

$$\begin{array}{r} 8 \overline{) 34} \\ 8 \overline{) 4} - 2 \end{array}$$

2) $54_{10} = 66_8$

$$\begin{array}{r} 8 \overline{) 54} \\ 8 \overline{) 6} - 6 \end{array}$$

3) $135_{10} = 207_8$

$$\begin{array}{r} 8 \overline{) 135} \\ 8 \overline{) 16} - 7 \\ 2 - 0 \end{array}$$

4) $345_{10} = 531_8$

$$\begin{array}{r} 8 \overline{) 345} \\ 8 \overline{) 43} - 1 \\ 5 - 3 \end{array}$$

5) $255_{10} = 377_8$

$$\begin{array}{r} 8 \overline{) 255} \\ 8 \overline{) 31} - 7 \\ 3 - 7 \end{array}$$

(4)

$$1) 64_8 = 52_{10} //$$

$$6 \times 8 + 4$$

$$2) 34_8 = 28_{10} //$$

$$3 \times 8 + 4$$

$$3) 77_8 = 63_{10} //$$

$$7 \times 8 + 7$$

$$4) 45_8 = 37_{10} //$$

$$4 \times 8 + 5$$

$$5) 13_8 = 11_{10} //$$

$$1 \times 8 + 3$$

(5)

$$1) 1256_{10} = 4F8_{16} //$$

$$\begin{array}{r} 16 \overline{) 1256} \\ 16 \overline{) 78} - 8 \\ 4 - 16 \end{array}$$

$$2) 56_{10} = 38_{16} //$$

$$\begin{array}{r} 16 \overline{) 56} \\ 3 - 8 \end{array}$$

$$3) 320_{10} = 140_{16} //$$

$$\begin{array}{r} 16 \overline{) 320} \\ 16 \overline{) 20} - 0 \\ 1 - 16 \end{array}$$

$$4) 48_{10} = 30_{16} //$$

$$\begin{array}{r} 16 \overline{) 48} \\ 3 - 0 \end{array}$$

$$5) 34_{10} = 22_{16} //$$

$$\begin{array}{r} 16 \overline{) 34} \\ 2 - 2 \end{array}$$

(6)

$$1) 1101111_2 = 337_8 //$$

$$\begin{array}{ccc} 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 3 & 3 & 7 \end{array}$$

$$2) 100000_2 = 40_8 //$$

$$\begin{array}{ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 4 & 0 \end{array}$$

$$3) 110001_2 = 141_8 //$$

$$\begin{array}{ccc} 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 4 & 1 \end{array}$$

$$4) 10011001_2 = 221_8 //$$

$$\begin{array}{ccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 2 & 2 & 1 \end{array}$$

$$5) 110001_2 = 61_8 //$$

$$\begin{array}{ccc} 1 & 1 & 0 & 0 & 0 & 1 \\ 6 & 1 \end{array}$$

(7)

$$1) 34_8 = 011100_2 //$$

$$2) 52_8 = 101010_2 //$$

$$3) 124_8 = 001010100_2 //$$

$$4) 22_8 = 010010_2 //$$

$$5) 23_8 = 010011_2 //$$

(8)

$$1) 11110000_2 = F0_{16} //$$

$$\begin{array}{ccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ F & 0 \end{array}$$

$$2) 11001000_2 = C8_{16} //$$

$$\begin{array}{ccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ C & 8 \end{array}$$

$$3) 1101010_2 = 6A_{16} //$$

$$\begin{array}{ccc} 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 6 & A \end{array}$$

$$4) 1110000_2 = 70_{16} //$$

$$\begin{array}{ccc} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 7 & 0 \end{array}$$

$$5) 10000_2 = 10_{16} //$$

$$\begin{array}{ccc} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 \end{array}$$

(9)

1) $2A_{16} = 00101010_2 //$

2) $349_{16} = 001101001000_2 //$

3) $23_{16} = 00100011_2 //$

4) $AA_{16} = 10101010_2 //$

5) $3D_{16} = 00111101_2 //$

(10)

1) $45_8 = 25_{16} //$

$$\begin{array}{r} 100101 \\ 2 \quad 5 \end{array}$$

2) $67_8 = 37_{16} //$

$$\begin{array}{r} 110111 \\ 3 \quad 7 \end{array}$$

3) $341_8 = E1_{16} //$

$$\begin{array}{r} 01110001 \\ E \quad 1 \end{array}$$

4) $234_8 = 9C_{16} //$

$$\begin{array}{r} 01001100 \\ 9 \quad C \end{array}$$

5) $23_8 = 13_{16} //$

$$\begin{array}{r} 010011 \\ 1 \quad 3 \end{array}$$

(11)

1) $2A_{16} = 52_8 //$

$$\begin{array}{r} 00101010 \\ 5 \quad 2 \end{array}$$

2) $34D_{16} = 1515_8 //$

$$\begin{array}{r} 00110101101 \\ 1 \quad 5 \quad 1 \quad 5 \end{array}$$

3) $50_{16} = 120_8 //$

$$\begin{array}{r} 01010000 \\ 1 \quad 2 \quad 0 \end{array}$$

4) $91A_{16} = 4432_8 //$

$$\begin{array}{r} 10010011010 \\ 4 \quad 4 \quad 3 \quad 2 \end{array}$$

5) $14_{16} = 24_8 //$

$$\begin{array}{r} 00101000 \\ 2 \quad 4 \end{array}$$